Orni at StanceEval-2026: Arabic Stance Detection with Reasoning-Augmented LLM Prompting and Encoder Ensembles

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Abstract

Arabic stance detection identifies whether an Arabic text expresses FAVOR, AGAINST, or NONE toward a given target. It is important for understanding public opinion and social media discourse, particularly in Arabic, where dialectal variation, sarcasm, and informal expressions make stance interpretation difficult. To address this, we took part in StanceEval-2026 Track 1, where models are trained on provided targets and evaluated on a related target from the same thematic domain. We investigate two complementary approaches: fine-tuning-based systems using Arabic encoders and parameter-efficient LLM adaptation, and prompted GPT-4o without fine-tuning. Our experiments include encoder ensembles, multi-task and contrastive learning, attention aggregation, augmentation, calibration, QLoRA, and prompted LLMs with few-shot demonstrations, target-specific context, structured reasoning, and self-consistency voting. The best fine-tuning-based system achieves 0.701 Favg2, while GPT-4o v4 reaches 0.827 Favg2, 0.630 Favg3, and 0.784 accuracy, ranking 11th on the official leaderboard. The results show that prompted GPT-4o transfers more effectively to the related-target setting, while additional prompt complexity does not always improve performance.

1 Introduction

Stance detection identifies whether a text expresses FAVOR, AGAINST, or NONE toward a specific target. This task is especially challenging in Arabic due to dialectal variation and the prevalence of informal language. Sarcasm further complicates the task. StanceEval-2026 (Albalawi et al., 2026a) utilizes the Mawqif-XT dataset (Albalawi et al., 2026b). In this work, we participate in Track 1, "Seen Targets," in which systems are trained on *COVID-19 Vaccine*, *Digital Transformation*, and *Women Empowerment* and subsequently evaluated on a related target within the same thematic domain.

Stance detection is a key tool for understanding public opinion at scale, supporting applications such as tracking attitudes toward health policies, monitoring social debates, and analyzing reactions to public campaigns. It is especially valuable for Arabic, one of the most widely used languages online, whose opinionated social-media content remains comparatively underexplored. This work explores various fine-tuning-based Arabic models with a prompted GPT-4o system. The fine-tuning-based approach incorporates encoder ensembles, multi-task and contrastive learning, attention aggregation, data augmentation, calibration, and QLoRA. The prompted GPT-4o system employs few-shot demonstrations, structured reasoning, and self-consistency voting.

The main contributions of this work are as follows:

- We systematically compare a comprehensive set of fine-tuning-based Arabic stance-detection methods with prompted large language models (LLMs) under a unified related-target evaluation setting.
- We develop a reasoning-augmented GPT-4o prompting strategy that integrates few-shot demonstrations, target-specific context, structured reasoning, and self-consistency voting.
- We provide empirical evidence demonstrating that prompting-based GPT-4o generalizes more effectively than fine-tuning-based approaches.

2 Background

2.1 Task Setup

Given a tweet t and a target g, the task is to predict a stance label $y \in \{\text{FAVOR}, \text{AGAINST}, \text{NONE}\}$ toward the target. The input consists of an Arabic tweet paired with a target phrase, and the output is one of the three stance classes. Stance is determined with respect to the target rather than the overall sentiment of the tweet. For example, a tweet supporting a target is labeled FAVOR, one opposing it is

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AGAINST, while a factual, ambiguous, or off-topic tweet is labeled NONE.

2.2 Dataset

We use Mawqif-XT (Albalawi et al., 2026b), an extension of the Mawqif Arabic stance corpus (Alturayeif et al., 2022), collected from X (Twitter). The dataset contains both Modern Standard Arabic and dialectal Arabic in short social-media posts containing phenomena such as hashtags, mentions, emojis, and informal spellings. Tweets are annotated for stance, with additional sentiment and sarcasm information.

Split	Fav	Agn	None	T_S	T_{UW}
Train	2,148	1,021	333	3,502	27,792
Dev	380	180	59	619	8,162
Test	—	—	—	352	3,123

Table 1: Class-wise distribution (Fav/Agn/None), total samples (T_S), and unique words (T_{UW}) per split. Test stance labels are withheld (blind).

The training and development sets contain the targets *COVID-19 Vaccine*, *Digital Transformation*, and *Women Empowerment*. As shown in Table 1, both the training and development sets are dominated by the FAVOR class (about 61%), with NONE being the rarest (under 10%). The training targets are predominantly FAVOR, whereas the blind test set has a substantially different label distribution, making generalization across targets a central challenge of the task. The code is publicly available.[1]

2.3 Related Work

Stance detection was first studied extensively for English, where early neural approaches such as bidirectional conditional encoding established the target-conditioned formulation of the task (Augenstein et al., 2016), and it has since been extended to many languages and to cross-target and multilingual settings. For Arabic, progress has been driven by the Mawqif corpus and the StanceEval shared task (Alturayeif et al., 2024). Arabic-specific pretrained encoders, notably MARBERTv2 (Abdul-Mageed et al., 2021) and AraBERT (Antoun et al., 2020), provide strong backbones for these systems by modelling both Modern Standard Arabic and dialectal social-media text. Building on them, several StanceEval-2024 systems used fine-tuned Arabic encoders enhanced with multi-task learning and contrastive objectives (Badran et al., 2024) or attention-based ensemble aggregation (Hasanaath and Alansari, 2024). These approaches motivate several of our fine-tuning-based configurations.

A related line of research considers cross-target and zero-shot stance detection, where models must generalize beyond the targets observed during training. Explicit reasoning has been explored for cross-target stance detection (Taranukhin et al., 2024), while self-consistency can improve LLM predictions by aggregating multiple reasoning outputs (Wang et al., 2023). Motivated by these directions, we compare fine-tuning-based Arabic systems with prompted LLMs using few-shot demonstrations, target context, structured reasoning, and self-consistency voting.

3 System Overview

We evaluate two complementary paradigms for Track 1: fine-tuning-based models and prompted GPT-4o. Both must transfer stance cues from the provided training targets to a related target rather than rely on target-specific lexical patterns. Figure 1 summarizes the methodology.

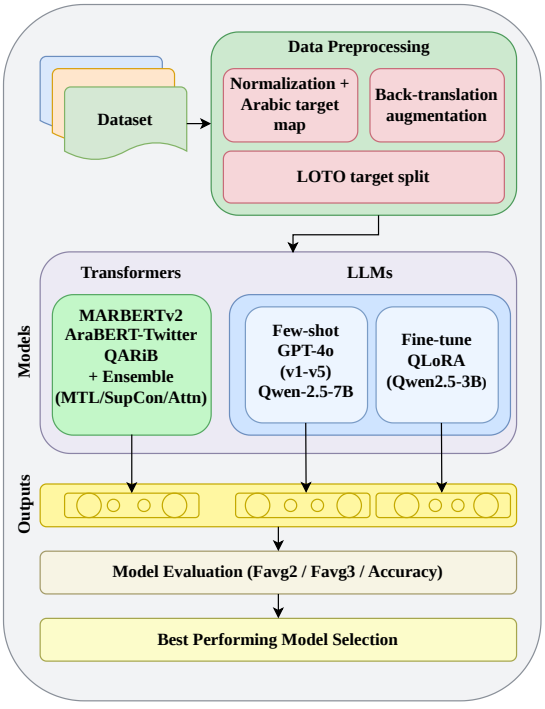


Figure 1: Overview of the proposed fine-tuning and prompted-LLM pipelines.

[1] https://github.com/DipannitaOrni/Orni-at-StanceEval-2026-ArabicNLP

3.1 Preprocessing and Resources

All fine-tuned models use the same normalization pipeline: URLs and mentions are replaced with Arabic placeholders, hashes and underscores are removed, Arabic diacritics and *tatweel* are stripped, and long character repetitions are collapsed. Fine-tuned encoder models use a maximum sequence length of 128 tokens, while QLoRA uses 160 tokens as detailed in Appendix A.1. English target names are mapped to Arabic.

For final fine-tuned systems, the development set is folded into training. Additional resources are limited to Mawqif sentiment and sarcasm annotations for multi-task learning, round-trip translation for augmentation, and the GPT-4o API for prompted inference. No external labeled stance data or stance/sentiment lexicons were used.

3.2 Fine-Tuning-Based Models

Our primary encoder ensemble averages MARBERTv2 (Abdul-Mageed et al., 2021) and AraBERT-Twitter (Antoun et al., 2020). Class imbalance is handled with inverse-frequency weighted cross-entropy,

$$w_c = \frac{N}{C \cdot n_c}, \tag{1}$$

with C=3 and label smoothing 0.05.

To estimate transfer to unseen targets, we use Leave-One-Target-Out (LOTO) validation over the three training targets. Final models are retrained on all labeled data with seeds 42 and 1234, and their probabilities are averaged. On LOTO out-of-fold predictions, we grid-search a narrow per-class multiplicative decision-bias vector that maximizes Favg2; wider ranges were found to overfit the training target distribution.

We additionally evaluate the following variants:

- **Encoder variants:** adding QARiB (Abdelali et al., 2021) and alternative two- and three-model combinations.
- **Multi-task learning:** joint stance, sentiment, and sarcasm heads with loss weights 1.0/0.5/0.3, following Badran et al. (2024); only stance is used at inference.
- **Supervised contrastive learning:** SupCon (Khosla et al., 2020) with τ=0.07 and weight 0.3.
- **Attention aggregation:** learned fusion of encoder [CLS] representations, following Hasanaath and Alansari (2024).
- **Back-translation:** Arabic→English →Arabic paraphrases using MarianMT (Tiedemann and Thottingal, 2020).
- **Distribution calibration:** probability rescaling toward the published test-target class distribution at several strengths.
- **QLoRA:** Qwen2.5-3B-Instruct (Yang et al., 2024), fine-tuned with 4-bit QLoRA (Dettmers et al., 2023) using rank 16, α=32, and dropout 0.05; its probabilities are also blended with the encoder ensemble.

3.3 Prompted GPT-4o Models

GPT-4o is used without fine-tuning. We submitted five configurations varying the number of demonstrations, reasoning structure, and voting strategy. Their settings are summarized in Table 4 in Appendix A.4, while the prompt structure and system instruction are given in Appendix A.5.

All versions use labeled demonstrations from the training targets. From v2, demonstrations follow a JSON schema with placeholder reasoning fields, and v2–v5 use self-consistency voting across multiple temperatures (Wang et al., 2023); v1 uses a single prediction.

Version v3 adds target-specific context. Version v4 introduces the three-step reasoning sequence *target_background*→*tweet_claim* →*stance_inference*, following Taranukhin et al. (2024). Version v5 adds harder demonstrations, more votes, and an optional encoder fallback.

4 Experimental Setup

We use the official Track 1 train, development, and blind test splits described in Section 2. Model selection and decision-bias calibration use Leave-One-Target-Out (LOTO) validation over the three training targets. For final fine-tuned systems, the training and development sets are merged into 4,121 tweets. Tweets are normalized by replacing URLs and mentions, removing Arabic diacritics and *tatweel*, and collapsing repeated characters. For encoder models, we encode each target–tweet pair up to 128 tokens. Encoders are trained for 4 epochs with batch size 16, learning rate 2×10^{-5}, AdamW, cosine scheduling with 10% warmup, label smoothing 0.05, gradient clipping at 1.0, and mixed precision. Final predictions average seeds 42 and 1234; detailed settings are given in Appendix A.1.

GPT-4o is accessed through the OpenAI API us-

ing English instructions with Arabic tweets and demonstrations. Version v1 uses a single prediction, while v2–v5 use self-consistency voting. Prompt configurations are summarized in Table 4, and the v4 prompt structure is given in Appendix A.5. We use PyTorch and HuggingFace Transformers[2] for encoder and QLoRA models, with PEFT[3] and bitsandbytes for 4-bit QLoRA, and scikit-learn for evaluation. Back-translation uses the Helsinki-NLP OPUS-MT models opus-mt-ar-en and opus-mt-en-ar[4], while prompted models use the OpenAI API[5] (gpt-4o).

Model checkpoints are obtained from the HuggingFace Hub, and encoder and QLoRA models are trained on a single NVIDIA T4 GPU. Official evaluation uses Favg2, Favg3, and accuracy; metric definitions are provided in Appendix A.2.

5 Results and Analysis

Table 2 reports the official Track 1 test-set results for all submitted system configurations. GPT-4o v4 achieved the best overall performance with Favg2 of 0.827, Favg3 of 0.630, and accuracy of 0.784.

System	Favg2	Favg3	Acc
Fine-tuning-based systems			
MARBERTv2 + AraBERT-Tw. (raw)	0.664	0.480	0.622
+ bias calibration	0.701	0.495	0.659
+ distribution calib.	0.670	0.486	0.616
+ CAMeLBERT (3-model)	0.660	0.479	0.622
+ QARiB (3-model)	0.658	0.488	0.619
MARBERTv2 + AraBERT-large-Tw.	0.682	0.497	0.645
MARBERTv2 + XLM-R-large	0.652	0.463	0.619
Multi-task learning	0.681	0.485	0.645
Supervised contrastive	0.696	0.510	0.651
Attention aggregation	0.671	0.480	0.622
Back-translation aug.	0.679	0.485	0.645
Pseudo-labeling	0.688	0.477	0.662
QLoRA (Qwen2.5-3B)	0.676	<u>0.537</u>	0.645
+ BERT blend	0.685	0.494	0.642
Prompted LLMs			
Qwen2.5-7B (few-shot)	0.528	0.426	0.480
GPT-4o v1	0.769	0.618	0.705
GPT-4o v2	0.803	0.625	0.747
GPT-4o v3	0.813	0.609	0.770
GPT-4o v4	**0.827**	**0.630**	**0.784**
GPT-4o v5	0.817	0.617	0.778

Table 2: Official Track 1 test results. Favg2 is the ranked metric. Best overall values are in **bold**; the best fine-tuning-based Favg3 is <u>underlined</u>.

Fine-tuning reaches a ceiling (∼0.70). Across fourteen fine-tuning-based configurations, Favg2 does not exceed 0.701 (Table 2). The two-model ensemble with decision-bias calibration performs best among these systems. Decision-bias calibration provides the largest improvement over the raw ensemble, increasing Favg2 from 0.664 to 0.701 and accuracy from 0.622 to 0.659. Other changes, including larger encoders, additional ensemble members, multi-task learning, supervised contrastive learning, augmentation, pseudo-labeling, and QLoRA, do not surpass this result. QLoRA obtains the highest Favg3 among fine-tuning-based systems (0.537), giving the strongest three-class average among these systems, although blending it with the encoder system does not improve Favg2.

Prompting surpasses the fine-tuning ceiling. GPT-4o with the basic v1 few-shot configuration reaches Favg2 of 0.769, already 0.068 above the best fine-tuning-based system. The later prompt configurations further improve performance: v2 reaches 0.803, v3 reaches 0.813, and the three-step reasoning configuration v4 reaches 0.827. GPT-4o v4 therefore improves over the best fine-tuning-based system by 0.126 Favg2 and achieves the best overall Favg2 (0.827), Favg3 (0.630), and accuracy (0.784).

Prompt improvements are not monotonic. The open Qwen2.5-7B few-shot baseline reaches Favg2 of only 0.528, substantially below GPT-4o under our prompting experiments, suggesting that base-model capability is an important factor in this task. Increasing the voting budget and introducing harder demonstrations in GPT-4o v5 does not improve over v4 (0.817 vs. 0.827). Thus, additional prompt complexity does not necessarily translate into better stance-detection performance.

6 Conclusion

This work presented our systems for StanceEval-2026 Track 1, comparing fourteen fine-tuning-based configurations with a prompted GPT-4o system. Fine-tuning plateaued near 0.70 Favg2, whereas GPT-4o with structured reasoning and self-consistency voting reached 0.827. The large gap over the Qwen2.5-7B baseline suggests base-model capability is a key factor. Future work will explore combining LLM and encoder predictions.

[2] https://github.com/huggingface/transformers
[3] https://github.com/huggingface/peft
[4] https://huggingface.co/Helsinki-NLP
[5] https://platform.openai.com

Limitations

This study has several limitations. First, the best-performing system relies on GPT-4o, a proprietary, paid model accessed via an API, which restricts reproducibility and long-term availability due to potential changes in the hosted model. Second, the prompted large language model (LLM) results are based on single leaderboard submissions without repeated runs, preventing the reporting of variance for these systems. Third, the evaluation focuses on a single related target (Women Driving) within one thematic domain, so the generalizability of these findings to other targets and domains has yet to be established. Finally, decision-bias calibration and few-shot demonstration selection are tuned on the training targets, which may not transfer effectively to targets with different stance distributions.

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A Appendix

A.1 Parameter Setting

Table 3 lists the full training configuration. All encoder variants share these settings; large models (AraBERT-large-Twitter, XLM-R-large) use a batch size of 8 to fit GPU memory. We trained the encoder and QLoRA models on a single NVIDIA T4 GPU (Kaggle); prompted GPT-4o runs require no GPU.

Setting	Value
Encoder fine-tuning	
Epochs	4
Batch size	16 (8 for large models)
Learning rate	2×10^{-5}
Optimizer	AdamW (weight decay 0.01)
LR schedule	cosine, 10% warmup
Max sequence length	128
Label smoothing	0.05
Gradient clipping	1.0
Precision	mixed (fp16)
Seeds	42, 1234
QLoRA (Qwen2.5-3B)	
Quantization	4-bit NF4, double quant
Compute dtype	bfloat16
LoRA rank / α	16 / 32
LoRA dropout	0.05
Target modules	all attn. + MLP proj.
Epochs	1–3
Learning rate	2×10^{-4}
Optimizer	paged AdamW 8-bit
Max sequence length	160
Prompted GPT-4o (v4)	
Model	gpt-4o
Few-shot examples	8 per class (24 total)
Self-consistency	5 votes
Temperatures	0.0, 0.2, 0.4, 0.6, 0.8
Max tokens	180
Response format	JSON object

Table 3: Full hyperparameters for all system families.

A.2 Evaluation Metric

The official scorer computes a per-class F_1 from precision and recall. For a class c,

$$F_1^{(c)} = \frac{2\,P_c\,R_c}{P_c + R_c}, \tag{2}$$

where P_c and R_c are the precision and recall for class c. The ranked metric averages the Favor and Against classes,

$$\text{Favg2} = \frac{F_1^{(\text{Favor})} + F_1^{(\text{Against})}}{2}, \tag{3}$$

and the secondary metric additionally includes None,

$$\text{Favg3} = \frac{F_1^{(\text{Favor})} + F_1^{(\text{Against})} + F_1^{(\text{None})}}{3}. \tag{4}$$

Favg2 excludes the None class, focusing evaluation on the Favor and Against stances.

A.3 Error Analysis

Because the test set is blind, we analyze errors on the development set (619 tweets with gold labels), running GPT-4o v4 with train-only demonstrations to avoid label leakage. The system reaches Favg2 0.880 and accuracy 0.845 on this set; the confusion matrix is shown in Figure 2. Two patterns are notable. First, the None class is hardest (F_1 0.51, versus 0.90 for Favor and 0.86 for Against): tweets that merely mention a target without a clear position are easily read as leaning one way. Second, sarcastic tweets are over-represented among errors (8.3% versus 4.2% of all tweets), suggesting that irony contributes to some stance confusions.

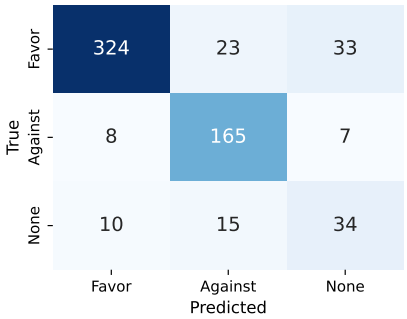


Figure 2: Confusion matrix on the development set for GPT-4o v4 (rows: gold labels, columns: predictions).

A.4 GPT-4o Prompt Configurations

Table 4 summarizes the five submitted GPT-4o configurations. The variants progressively increase the number of few-shot demonstrations and voting samples, while later versions introduce more structured reasoning.

Ver.	Reasoning	Shots	Votes
v1	None	9	1
v2	1 sentence	12	3
v3	1 sentence	24	5
v4	3-step	24	5
v5	3-step + hard ex.	24	9

Table 4: Submitted GPT-4o prompt configurations.

A.5 Prompt Structure and Design

The best system, GPT-4o v4, uses a structured inference prompt consisting of four components: a fixed system instruction, a target-specific context, 24 few-shot demonstrations, and the final target–tweet query. The demonstrations contain eight examples from each stance class (FAVOR, AGAINST, and NONE) and are sampled from the labeled data using a fixed random seed of 42. In each demonstration, the three reasoning fields contain placeholders while the gold stance label is provided. Consequently, the demonstrations specify the output schema and label space, whereas the reasoning for the query instance is generated at inference time.

Table 5 shows the fixed system instruction used in GPT-4o v4, while Table 6 summarizes the complete inference-prompt structure. The three-step reasoning decomposition follows the explicit-reasoning approach of Taranukhin et al. (2024).

System Prompt (GPT-4o v4)

You are an expert annotator for Arabic (incl. dialectal/Gulf) stance detection. Given a TARGET and a TWEET, decide the tweet's stance TOWARD THE TARGET.

- **Favor**: supports/defends/promotes the target, INCLUDING by mocking or attacking the target's opponents.
- **Against**: opposes/criticizes/rejects the target, INCLUDING by mocking its supporters.
- **None**: mentions the topic but no clear position; purely factual/news; a question; ambiguous; off-topic.

Rules: stance is about the TARGET, not general emotion. Sarcasm/irony usually flips surface meaning—infer the author's true position. Religious/cultural appeals for or against count as stance. When genuinely unclear or purely descriptive, choose None.

Reason in THREE explicit steps, then answer:
1. target_background: one sentence on what is being debated about the target.
2. tweet_claim: one sentence on what the tweet itself actually asserts, implies, or mocks.
3. stance_inference: one sentence connecting the claim to the target to justify the label.

Strict JSON:
{"target_background":"<...>",
"tweet_claim":"<...>",
"stance_inference":"<...>",
"stance":"Favor|Against|None"}

Table 5: Fixed system instruction used in GPT-4o v4. At inference time, it is combined with target-specific context, 24 few-shot demonstrations, and the final target–tweet query.

Component	Description
System instruction	Defines FAVOR, AGAINST, and NONE; specifies the three-step reasoning procedure; and requires a strict JSON response.
Target context	Provides a short target-specific description of the debate and clarifies how FAVOR and AGAINST should be interpreted for the current target.
Few-shot demonstrations	Contains 24 labeled examples: eight FAVOR, eight AGAINST, and eight NONE. The three reasoning fields contain placeholders, while the stance field contains the gold label.
Final query	Provides the target and the Arabic tweet to be classified.

Table 6: Structure of the GPT-4o v4 inference prompt.

A.6 Prediction Examples

Table 7 presents representative misclassified development-set examples from GPT-4o v4. The examples illustrate errors involving ambiguous stance, sarcasm, and target-specific interpretation.

Text Sample	Actual Label	Predicted Label
Sample 1: Target: *Digital Transformation* ممكن تفهمونا شنو التحول الالكتروني لو هو مجرد خبر *(Can you explain what digital transformation is, or is it just news?)*	None	Against
Sample 2: Target: *COVID-19 Vaccine* لانريد تطعيم والسلام مافي عندنا شي يعني بالصميل نأخذ لقاح *(We do not want vaccination; are we supposed to take the vaccine by force?)*	Favor	Against
Sample 3: Target: *Women Empowerment* #المواشي_سوق تمكين المرأة من دخول بيع وشراء في سوق الاغنام. #البيئة_وزارة اغنام كيوت URL *("Empowering" women to enter buying and selling in the sheep market. "Cute sheep.")*	Against	Favor
Sample 4: Target: *Digital Transformation* #الالكتروني_للحب_كلمه عجي ما عاد تتحمل التعامل مع الحب الالكتروني لانها ضعيفه وبثها مو ثابت يجب التحول فورا لخدمه ٥ جي لمواكبة السرعه URL *(4G can no longer handle electronic communication reliably; we should switch immediately to 5G to keep up with the speed.)*	Favor	None

Table 7: Representative misclassified examples showing the Arabic tweet, English gloss, actual stance label, and model prediction.